

Shashank Suman



LinkedIn



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ResearchGate



Webpage

Trieste, Italy | (+91) 9523815435 (Personal)

Education and Training

Five-Year Integrated Masters in Basic Sciences (Physics)

UM-DAE Centre for Excellence in Basic Sciences

August 2018 - May 2023

Mumbai, India

Relevant Coursework: Electromagnetism, Quantum Mechanics, Mathematical Methods, Condensed Matter Physics, Atomic and Molecular Physics, Statistical Mechanics, Electronics and Computational Techniques

Publications

S. Suman, E. Bashaiah, M. Resmi and R. Yalla “*In situ* characterization of optical micro/nano fibers using scattering loss analysis”, J. Appl. Phys. 135, 123101 (2024) DOI: <https://doi.org/10.1063/5.0192385>

E. Bashaiah, **S. Suman**, M. Resmi, B. Das and R. Yalla, “Fabrication and characterization of optical micro/nanofibers”, Journal of Nanophotonics 18(3), 036007 (2024) DOI: <https://doi.org/10.1117/1.JNP.18.036007>

M Resmi, E. Bashaiah, **S. Suman** and R. Yalla, “Highly efficient coupling of single photons using a pair of nanostructures”, Opt Quant Electron 56, 1341 (2024). DOI: <https://doi.org/10.1007/s11082-024-07256-z>

Work Experience

TRIL Fellow | Quantum Interconnects with cold atoms

with Dr. Matteo Marinelli, University of Trieste

October 2025 -

Trieste, Italy

- * Characterisation of ILA and Optical Isolator using interferometer set-up
- * Designing and fabricating PCB circuit for low current variable gain photodiode
- * Coding the Control System for the experiment using ARTIQ and ATOMIQ
- * Developing driver for Mako-G series camera for faster initialisation using Vmbpy python package
- * Coding the LABJACK for monitoring lasers using python
- * Designing modules for holding and placing various components like photodiodes and distribution circuits

Visiting Research Student | Ultra Cold Rydberg Atoms

with Dr. Sanjukta Roy, Raman Research Institute

July 2024 - January 2025

Bengaluru, India

- * Calibration of helmholtz coil using Gauss probe for defining quantization axis in cold atom experiments
- * Designing the set-up for optical dipole traps using OSLO software
- * Realization of the experimental setup for measuring the line width of lasers (780 nm [DL 100]) used as probe and cooling beams
- * Focal spot measurement of 780 nm and 852 nm ($\approx 3 \mu\text{m}$) for dipole trap using knife edge technique and an interferometer
- * Optimization of Magneto-optical trap parameters for efficient loading of optical dipole traps
- * Realization and optimization of PID circuit to stabilize intensity fluctuations of dipole laser after AOM
- * Realization of Saturated Absorption Spectroscopy (SAS) setup using Caesium transition line and locking 852 nm laser using the same
- * Realisation of the optical dipole trap setup with imaging and correlation system
- * Locking of multiple lasers (960 nm [TASHG Pro 960] and 780 nm [DL 100]) simultaneously using Ultra-Low Expansion (ULE) transfer cavity

CHANAKEYA-PG Fellow | Optical Micro/Nano Fibres and Tips

with Dr. Ramachandrarao Yalla, University of Hyderabad

May 2023 - June 2024

Hyderabad, India

- * Simulating the interaction of nanoparticles of various sizes with tapered fibre regions of different diameters
- * Simulating wavelength coupling and enhancement for biosensing and quantum networks
- * Conducted sensing and sizing experiments using optical micro/nanofibers (OMNFs)
- * Designing the experimental set-up for verification of simulations and conducting relevant experiments
- * Coordinating biosensing and plasmonic-enhanced coupling experiments
- * Fabrication of OMNFs using heat and pull technique using small hydrogen flame torch
- * Assisted in the setup and re-establishment of the laboratory from the ground up after relocation

Master's Thesis Student | Optical Micro/Nano Fibres

with Dr. Ramachandrarao Yalla, University of Hyderabad

July 2022 - December 2022

Hyderabad, India

- * Fabricated and optimized chemically etched OMNFs

- * Characterized OMNFs using conventional methods like *field emission scanning electron microscopy (FESEM)* and *energy-dispersive X-ray analysis (EDAX)*
- * Developed a *non-destructive and an in-situ characterization technique* for fabricated OMNF
- * Theoretical analysis of *modes and photonic interactions*

Visiting Project Student | Vacancy Centres in Diamonds
with [Dr. Sushil Mujumdar](#), TIFR and [Dr. Padmnabh Rai](#), CEBS

February 2022 - May 2022
Mumbai, India

- * Acquired in-depth knowledge of the *theory of colour centres in diamonds*
- * Designed and optimized a *confocal microscope* for single-photon emission studies
- * Optimized parameters for *ion implantation on diamond wafer surfaces*

Projects

10th Semester Project | Time Crystals
with [Dr. Aameya A. Bhagwat](#), UM-DAE CEBS

January 2023 - April 2023
Mumbai, India

- * Acquired comprehensive understanding of *fundamental concepts and historical development* of time crystals
- * Analysed *future prospects and limitations* of current advancements
- * Monitored the progress in *experimental realization* of these crystals

Distributed Quantum Networks
with [Dr. Padmnabh Rai](#), UM-DAE CEBS

January 2022
Mumbai, India

- * Reviewed and discussed key aspects of *quantum networks*

Single Photon Emitters
with [Dr. Sushil Mujumdar](#), TIFR

January 2022
Mumbai, India

- * Conducted an in-depth study of *various potential quantum sources of photons* and the underlying physics

Semester Project | Quantum Information Science
with [Prof. Sudhir R. Jain](#), UM-DAE CEBS

August 2021 - December 2021
Mumbai, India

- * Gained a thorough *understanding of the mathematics* behind quantum information science
- * Studied the *fundamentals of quantum error correction* theory
- * Conducted a brief overview of *quantum processor hardware*

Conferences and Workshops

- * **International school on Frontiers of Light- Hot topics in Photonics** *October 2023 | Hyderabad, India*
ICFO-TIFR, Hyderabad.
Presented a poster titled, “**In-situ characterisation of optical micro/nano fibres**”
- * **Indo-French Conference on Frontiers in Photonics and Metamaterials-2023** *April 2023 | Hyderabad, India*
Mahindra University, Ecole Centrale School of Engineering
Presented a poster titled, “**Fabrication and Analysis of tapered micro/nano fibres for sensing applications**”
- * **Frontiers in Physics-2023** *March 2023 | Hyderabad, India*
University of Hyderabad
Presented a poster titled, “**Fabrication and Analysis of tapered micro/nano fibres for sizing applications**”
- * **Summer School on Quantum Information and Quantum Technology-2023** *June 2023 | Kolkata, India*
IISER-Kolkata (online)
- * **Summer School on Quantum Information and Quantum Technology-2022** *June 2022 | Kolkata, India*
IISER-Kolkata (online)
- * **Quantum Information to Application** *October 2021 | Jodhpur, India*
IIT-Jodhpur (online)

Research Interest

Interested in exploring fields that utilizes *Light-Matter Interactions* as basis for better understanding topics like *many-body physics, time crystals and new age computing hardware*.

Skills

- * **Programming:** Python(Qiskit, SciPy, Matplotlib, Numpy, Pandas), R, FORTRAN, C++, Java
- * **Software:** LINUX, LabVIEW, Mathematica, MATLAB, Gnuplot, OriginPro, ImageJ, Ansys LUMERICAL, OSLO, Adobe Illustrator

- * **Hardware:** OMNF fabricating module, Sensitive optical power meters, Single photon counting module, FESEM, EDAX, APD, Sensitive spectrometers, Liquid-cooled EMCCD cameras, CMOS cameras, AFM, XRD, Control systems for Ion accelerators, Nano/micro positioning stages, Piezo-electric stages, Vacuum Systems, External Cavity Tunable Mode locked Lasers, AOM, EOM, Spectrum Analyser, RF signal Generator, Photodetector (Broadband and Fast response), Control Modules of tunable lasers (Digital and Analog), Gauss Probe, Beam Profiler.
- * **Soft skills:** Problem-solving, Analytical Thinking, Teamwork, Effective communication, Time-management, Adaptability and Attention to detail.

Achievements

- * **CHANAKYA-PG** Fellowship by DST, Govt. of India May 2023 - November 2023
- * **DISHA** scholarship by DAE, Govt. of India 2018-2023
- * Secured a place **among top 1%** in National Entrance Screening Test and in Joint Entrance Exams (JEE)-Mains 2018

Miscellaneous

- * **Hobbies:** Music (vocal and instrumental), Poem Writing, Philosophical discussions, Sports and Dance
- * **Self Taught Subjects:** Fiber optics, Photonics and Nano-optics, Ultracold Atoms, Machine Learning and Introductory Quantum Information Science
- * **Management and Leadership:** Vice-president of Arts Society of CEBS 2019-20
- * **Teaching:** Been as a tutor and physics subject expert on [Chegg.com](https://www.chegg.com) since 2021
- * **Guidance:** Helped supervise short term bachelors and masters level projects in the lab 2023-2024

References

- * Dr. Matteo Marinelli [✉ matteo.marinelli@units.it](mailto:matteo.marinelli@units.it)
Assistant Professor, University of Trieste *TRIL supervisor*
- * Dr. Ramachandrarao Yalla [✉ rrysp@uohyd.ac.in](mailto:rrysp@uohyd.ac.in)
Assistant Professor, University of Hyderabad *Master's thesis guide, Project guide*
- * Dr. Sanjukta Roy [✉ sanjukta@rri.res.in](mailto:sanjukta@rri.res.in)
Associate Professor, Raman Research Institute, Bengaluru *Post Masters Project guide*
- * Dr. Padmnabh Rai [✉ padmnabh.raai@cbs.ac.in](mailto:padmnabh.raai@cbs.ac.in)
Reader(F), UM-DAE CEBS, Mumbai *Project guide, 2 theory courses and 1 lab course instructor*
- * Dr. Ameeya A. Bhagwat [✉ ameeya@cbs.ac.in](mailto:ameeya@cbs.ac.in)
Associate Professor, UM-DAE CEBS, Mumbai *Project guide and 2 theory courses instructor*